Cynthia Bortolotto

Postdoctoral Researcher, IMPA

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Research interests

- Mathematical analysis, particularly approximation theory and the mathematical foundations of neural networks.
- Lean's Mathlib library retrieval and efficient and accessible LLM-assisted formalization in Lean.

Academic appointments

- **Postdoctoral Researcher**, IMPA, Rio de Janeiro — April 2025–present.
- **Postdoctoral Researcher**, ETH Zürich — 2024–February 2025.

Education

- **Ph.D. in Mathematics**, ETH Zürich, 2020–2024. Advisor: Emmanuel Kowalski; co-advisor: Özlem Imamoglu. Thesis: *Distribution problems in analysis and number theory.*
- **M.Sc. in Mathematics**, IMPA, 2018–2020. Advisor: Emanuel Carneiro. Project: *The sphere packing problem in dimensions 8 and 24: an overview of the solution.*
- **Teaching Degree in Mathematics**, UNISINOS, 2015–2017. Advisor: Rogério Ricardo Steffenon. Final project: *Dirichlet L-functions and some results about prime numbers.*

Awards, honours and funding

- **FAPERJ Nota 10 Scholarship**, 2019. Awarded to the two top-performing first-year master's students at IMPA.
- **CAPES M.Sc. Fellowship**, 2018–2019.
- **CAPES OBMEP Scholarship**, 2015–2016. Scholarship for teachers and undergraduate teaching-degree students to provide mathematical olympiad training for award recipients.

Technical skills

Programming: Python. **Formal theorem proving:** Lean.

Machine Learning Projects

Centro Pi, IMPA.

- **Building Resilient Research in the Intelligence Age — OpenAI.** September 2026–present. Investigate the infrastructure, training and costs needed for sustainable adoption of AI in mathematical research. Construct a harness for LLMS that keeps the mathematician in the loop, prioritizing their input and supporting their understanding.
- **Complaint analysis — LabSeguros / Susep.** March 2026–present. Work on complaint and sentiment analysis using large language models (LLMs) and embedding models.
- **Deforestation monitoring — Imazon.** December 2025–February 2026. Studied potential active learning approaches for detecting deforestation in satellite imagery during the project's final phase.
- **Answer-sheet reading — OBMEP.** May–October 2025. Contributed to a project on automated reading of OBMEP answer sheets.

- **Embedding-based retrieval for Lean and mathlib.** Early-stage work exploring the fine-tuning of an embedding model to retrieve mathlib statements relevant to a given Lean statement, with the aim of finding related results or statements that imply the query.

Talks

- Presentation of the project *Análise de Sentimentos.* 1st Technical-Scientific Symposium of Instituto IRB, Porto Maravalley, Rio de Janeiro, Brazil, 10 September 2026.
- *Modular forms and Irrationality Proofs.* Algebra Group Seminar, ICMC–USP, 22 May 2026.
- *Modular forms and irrationality proofs.* II Semana de Encontros Aritméticos (II SemEAr), IMECC–UNICAMP, Campinas, Brazil, 26 February 2026.
- *Intersection of random chords in the circle.* Harmonic Analysis and Applications session, 35th Colóquio Brasileiro de Matemática, IMPA, Rio de Janeiro, Brazil, 29 July 2025.
- *Intersection of random chords and the Dilogarithm.* Geometry Seminar, IMPA, 27 May 2025.
- *Weyl Sums with Multiplicative Coefficients and Joint Equidistribution.* Analysis Seminar, UFRGS, 12 May 2025.
- *Intersection of random chords and the Dilogarithm.* Analysis Seminar, Universidade de Coimbra, 24 January 2025.
- *Intersection of random chords and the Dilogarithm.* Young Researchers Seminar, Instituto Técnico Superior, 13 December 2024.
- *On distribution of random chords in the plane.* Geometry Graduate Colloquium, ETH Zürich, 24 October 2024.
- *Prime time mysteries: a 165-year-old problem.* STEM For All Seminar, Minerva University, 13 September 2024.
- *Exponential Sums with Multiplicative Coefficients.* Number Theory Seminar, Institut Élie Cartan de Lorraine, 21 March 2024.
- *Weyl Sums with Multiplicative Coefficients and Joint Equidistributions.* Rencontres de théorie analytique et élémentaire des nombres, Institut Henri Poincaré, 18 March 2024.
- *Joint Equidistributions of congruence polynomial roots and polynomial values.* Colóquio Brasileiro de Matemática, 25 July 2023.
- *Weyl Sums with Multiplicative Coefficients and Joint Equidistributions.* Virtual Brazilian Analytic Number Theory Seminar, 31 January 2023.
- *What is sphere packing?* Graduate Colloquium, ETH Zürich, 10 May 2022.
- *O problema do empacotamento de esferas.* Seminário A5, IME–USP, 13 July 2021.
- *The sphere packing problem in dimensions 8 and 24* (jointly with T. Borges). Summer School on Sphere Packings and Optimal Configurations, 1 October 2019.
- *The sphere packing problem and the construction of magic functions* (jointly with T. Borges). ICTP Mathematics Seminar, 24 September 2019.
- *Endpoint Behaviour for a Maximally Modulated Averaging Operator* (jointly with T. Borges). Atelier d'Analyse Harmonique, 28 August 2019.
- *Endpoint Behaviour for a Maximally Modulated Averaging Operator* (jointly with T. Borges). IMPA Students' Colloquium, 23 August 2019.

Thesis Committees

- Member of the master's thesis examination committee for Henrique Sotel Pio, UFRGS, 3 March 2026. Thesis: *Aplicações das formas modulares à teoria de números.*
- Member of the master's thesis examination committee for Alexandre Dieguez, UFMG, 23 February 2026. Thesis: *Sobre certas correlações no problema do divisor de Dirichlet.*

Teaching and academic organization

- **2025.2** — Main instructor, *Medida e Integração*, IMPA, master's programme.
- **2025.1** — Main instructor, *Análise Complexa*, IMPA, master's programme.
- **2024.2** — Course coordinator, *Analysis I*, ETH Zürich. Main instructor: Andreas Steiger.
- **2024.1** — Course coordinator, *Analysis II*, ETH Zürich. Main instructor: Menny Akka Ginosar.
- **2023.2** — Course coordinator and teaching assistant, *Additive Combinatorics*, ETH Zürich. Main instructor: Emmanuel Kowalski.
- **2023.1** — Teaching assistant, *Analysis II*, ETH Zürich. Main instructor: Andreas Steiger.
- **2022.2** — Course coordinator and teaching assistant, *Complex Analysis*, ETH Zürich. Main instructor: Emmanuel Kowalski.
- **2022.1** — Teaching assistant, *Functional Analysis II*, ETH Zürich. Main instructor: Marc Burger.
- **2021.2** — Course coordinator, *Funktionentheorie*, ETH Zürich. Main instructor: Thomas Willwacher.
- **2021.1** — Course coordinator, *Sphere Packings, Lattices and Codes*, ETH Zürich. Main instructor: Danylo Radchenko.
- **2020.2** — Teaching assistant, *Análise no* R^n, IMPA. Main instructor: Roberto Imbuzeiro.

Publications

Peer-reviewed journal articles (4)

1. C. Bortolotto and V. Souza. Intersections of random chords of a circle. *The American Mathematical Monthly*, published online 20 May 2026. arXiv:2504.12866.
2. C. Bortolotto and J. P. G. Ramos. On extremal problems associated with random chords on a circle. *Mathematika* **71**(4), e70024, 2025. arXiv:2406.06771.
3. M. Bordignon, C. Bortolotto, and B. Kerr. Weyl sums with multiplicative coefficients and joint equidistribution. *Algebra & Number Theory* **19**(8), 1549–1580, 2025. arXiv:2303.03768.
4. T. Borges, C. Bortolotto, and J. P. G. Ramos. On the endpoint behaviour of oscillatory maximal functions. *Journal of Mathematical Analysis and Applications* **495**(1), 124670, 2021. arXiv:2001.11546.

Preprints (2)

1. C. Bortolotto and J. P. G. Ramos. *A complex-analytic proof of square-restricted stable phase retrieval in Fock space.* arXiv:2608.26365, 2026.
2. C. Bortolotto and L. Oliveira. *Modular Forms and Numerical Explorations of Rational Approximations to* $\zeta(3)$. arXiv:2605.00673, 2026.

Conferences, workshops and summer schools attended

- International Conference on Learning Representations (ICLR), 23–27 April 2026.
- II Semana de Encontros Aritméticos (II SemEAr), IMECC–UNICAMP, Campinas, Brazil, 24–27 February 2026.
- 35th Colóquio Brasileiro de Matemática, IMPA, Rio de Janeiro, Brazil, 28 July–1 August 2025.
- Fourier Analysis and Beyond, IMPA, Rio de Janeiro, Brazil, 30 June–4 July 2025.
- Colóquio Brasileiro de Matemática, Rio de Janeiro, Brazil, 24–28 July 2023.
- Group Actions and Harmonic Analysis in Number Theory, Oberwolfach, Germany, 7–12 May 2023.
- PCMI Graduate Summer School in Computational Tools for Number Theorists, Park City, USA, 18 May–5 August 2022.
- Young Scholars in the Analytic Theory of Numbers and Automorphic Forms, Bonn, Germany, 28–29 March 2022.

- Counting Problems, Ventotene, Italy, 6–11 September 2021.
- Harmonic Analysis and Analytic Number Theory Trimester Programme, HIM, Bonn, Germany, 3–26 June 2021.
- Online Summer School on Optimization, Interpolation and Modular Forms, 24–28 August 2020.
- Summer School on Sphere Packings and Optimal Configurations, Kopp, Germany, 29 September–4 October 2019.
- Atelier d'Analyse Harmonique: Masterclass in Harmonic Analysis, Saint-Nazaire, France, 27–30 August 2019.
- 32nd Colóquio Brasileiro de Matemática, IMPA, Rio de Janeiro, Brazil, 28 July–2 August 2019.
- Encontro Nacional de Mulheres Matemáticas, IMPA, Rio de Janeiro, Brazil, 27–28 July 2019.
- International Congress of Mathematicians, Rio de Janeiro, Brazil, 1–9 August 2018.
- World Meeting for Women in Mathematics, Rio de Janeiro, Brazil, 31 July 2018.
- ICM Satellite Conference on Harmonic Analysis, Porto Alegre, Brazil, 24–29 July 2018.

Additional information

- **Languages:** Portuguese (native), English (fluent), German (B1).
- **Citizenship:** Brazilian and Italian.